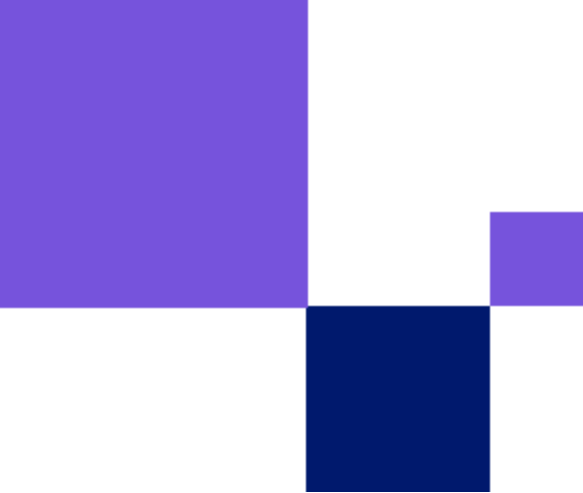




Foundations of Financial Accounting (ACC107)

Lecture 2
Double-entry Bookkeeping System
(Ch.9)

Learning outcomes

- Explain the principles of double-entry bookkeeping, including the purpose of having different ledger accounts.
- Describe the format and contents of the general ledger and ledger accounts.
- List the three steps to take when performing double entry.
- Distinguish between asset and expense accounts and capital, liability and income accounts.
- Record cash (including bank) transactions and credit transactions.

How to record business transactions?

Transaction:
The owner started a
business with £10,000 cash

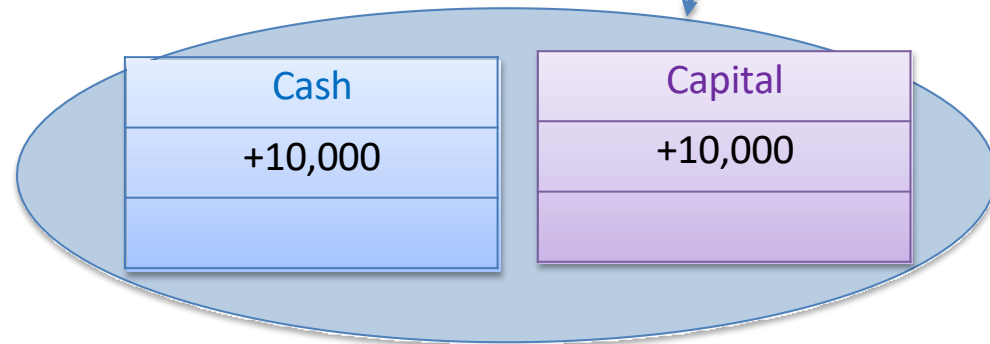
Single-entry bookkeeping system

details	Cash
Investment	+10,000

Our focus

Double-entry bookkeeping system

At least two accounts are affected



Ledger and accounts

Businesses **record transactions** with a book (or computerized system) called the **ledger**(分类账簿).

The ledger is divided into sections called **accounts** (账户). There is usually an 'account' for every class of expenditure, income, asset, liability, and capital.

Each account is split into two halves: the **left half** is called the **debit side** (借方). and the **right half** is called the **credit side** (贷方).

Debit (Dr.)/Credit (Cr.) just means increase (+) and decrease (-) for different sets of accounts.

Debit				Credit			
Date	Details	Folio	Amount	Date	Details	Folio	Amount
Cash account (page 1)							
6 Jan	Sales	p. 2	£100				

Debit				Credit			
Date	Details	Folio	Amount	Date	Details	Folio	Amount
Sales account (page 2)							
				6 Jan	Cash	p. 1	£100

Double-entry

Double-entry bookkeeping ensures that for every entry into an account, there needs to be a corresponding and opposite entry into a different account. It will result in a debit entry in one or more accounts and a corresponding credit entry in one or more accounts.

When recording transactions, at least two accounts are affected with double-entry bookkeeping.

The double-entry system of bookkeeping standardizes the accounting process and improves the accuracy of prepared financial statements, allowing for improved detection of errors.

T-account

T-account represents a simple version of the **ledger**.

Debit side			Account title		Credit side	
Cash account						
20X0	Details	£	20X0	Details	£	
1 Jan	Capital	1,000	1 Jan	Rent	100	
1 Jan	Loan – London Bank Ltd	500	2 Jan	Fixtures and fittings	300	
31 Jan	Sales revenue	600	8 Jan	Purchases	400	
			9 Jan	Carriage inwards	25	
			10 Jan	Stationery	50	
			15 Jan	Wages	200	
			20 Jan	Drawings	150	

Explanation: the **other** account that has been affected by the same transaction.

More simplified
version:

ACCOUNT TITLE	
£ XX	£ XXX
Debit (Left Side)	(Right Side) Credit

Accounting equation and T-account

► Recall accounting equation

Assets (A) = Capital (C) + Liabilities (L)

*Closing Assets (CA) = Opening Capital (OC) + New Capital (NC) – Drawings (D)
+ Revenue (R) – Expenses (E) + Closing Liabilities (CL)*



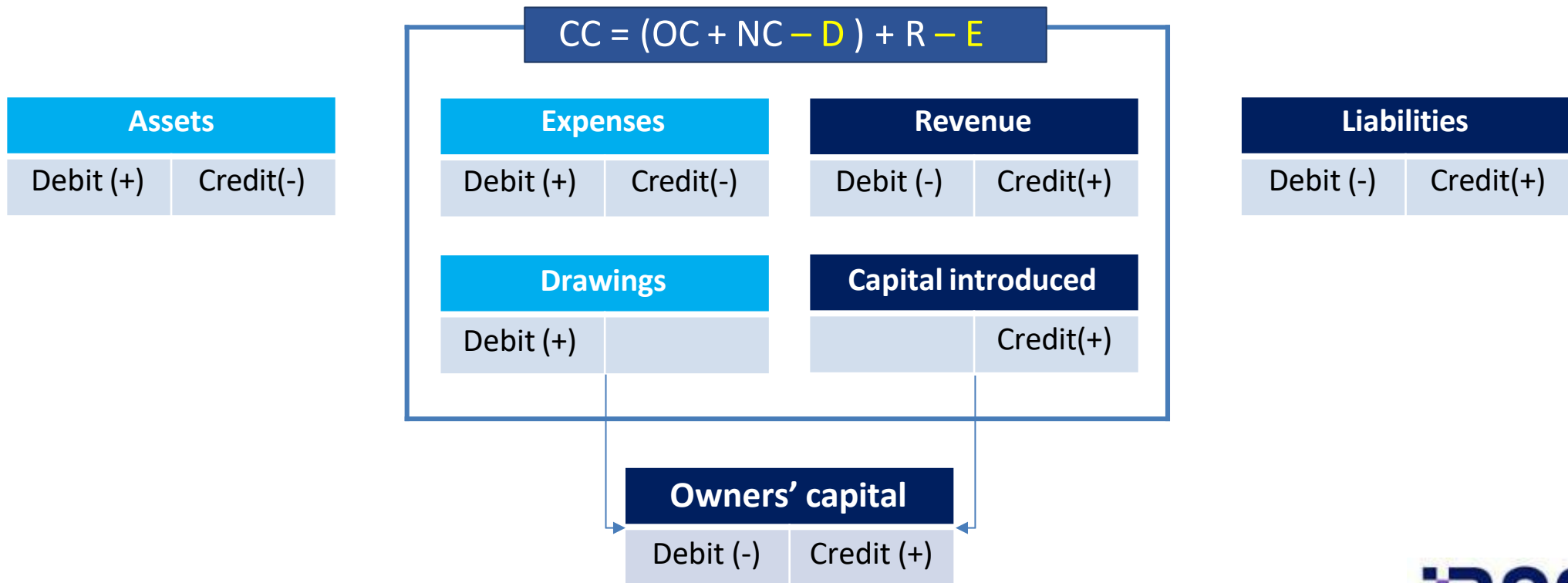
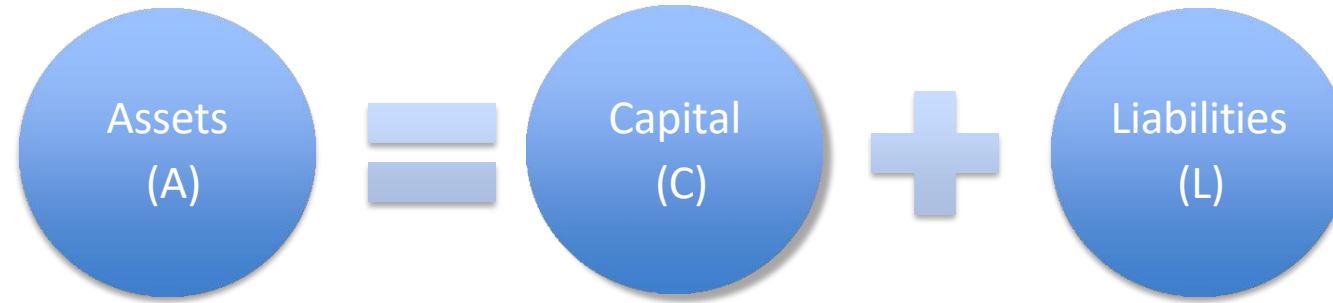
**Closing Assets (CA) + Drawings (D) + Expenses (E) = Opening Capital (OC) +
New Capital (NC) + Revenue (R) + Closing Liabilities (CL)**

$$\mathbf{A + D + E = C + R + L}$$

increase on their debit (left) side
decrease on their credit (right) side

increase on their credit (right) side
decrease on their debit (left) side

Accounting equation and T-account



*Drawings lead to a decrease in owners' capital.

Journal entry

Business transactions can also be recorded in journal form. Each entry in the journal consists of the following:

- the name of the ledger account that is to be debited and the amount
- the name of the ledger account that is to be credited and the amount

Example: The business sold goods amounting to £10,000 for cash on Jan 6

Jan 6	Dr.	Cash	10,000	
	Cr.	Sales revenue		10,000

T-account entry

Example: The business sold goods amounting to £10,000 for cash on Jan 6

Cash a/c						Sales a/c					
Date	Detail	Amount	Date	Detail	Amount	Date	Detail	Amount	Date	Detail	Amount
Jan 6	Sales Rev.	10,000							Jan 6	Cash	10,000

The diagram illustrates the flow of the T-account entry. A light blue arrow points from the 'Sales Rev.' entry in the Cash T-account to the 'Dr. Cash' entry in the journal entry table. A dark blue arrow points from the 'Cash' entry in the Sales T-account to the 'Cr. Sales revenue' entry in the journal entry table.

Jan 6	Dr.	Cash	10,000	
	Cr.	Sales revenue		10,000

Tips

► Three steps to record transactions in T-account or journal

1. Determine the 2 (or more) accounts affected.
2. Identify which category do the accounts belong to (A/L/C/R/E)
3. Consider the flow of value (increase or decrease; Dr. or Cr.).

Working example

* XYZ Ltd. Opened business on Jan 1, 2024. And the owner introduced a **cheque** (支票) of £8,000 into the business.

1

Bank a/c						Capital introduced a/c					
Date	Detail	Amount	Date	Detail	Amount	Date	Detail	Amount	Date	Detail	Amount
Jan 1	Cap. Intro.	8,000							Jan 1	Bank	8,000

2

Jan 1	Dr.	Bank (*Asset ↑)	8,000	
	Cr.	Capital introduced (*Capital ↑)		8,000

3

Check the accounting equation:

A (8,000) = C (8,000) + L (0)

Recall: Accounting is a business language!

1 = 2 = 3

*Account names will be provided in the quizzes and exams.

Working example

* Jan 3 - Purchased goods for £2,500 for resale by cheque.

Bank a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 1	Cap. Intro.	8,000	Jan 3	Pur.	2,500

Capital introduced a/c					
Date	Detail	Amount	Date	Detail	Amount
			Jan 1	Bank	8,000

Purchase a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 3	Bank	2,500			

*A **purchase** account is used to record the **expenses** related to the purchases of goods for resale under the periodic inventory system. (seemore in Lecture 4)

Jan 3	Dr.	Purchase (*Expense ↑)	2,500	
	Cr.	Bank (*Assets ↓)		2,500

Check the accounting equation:

Assets: Bank $(8,000 - 2,500 = 5,500)$

Capital: (OC + NC – D) + R – E $= (8,000 + 0 - 0 + 0 - 2,500 = 5,500)$

Liability: Nil

A (5,500) = C (5,500) + L (0)

Working example

* Jan 20 - Sold the goods **on credit** to Doyle for £3,400.

Bank a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 1	Cap. Intro.	8,000	Jan 3	Pur.	2,500

Capital introduced a/c					
Date	Detail	Amount	Date	Detail	Amount
			Jan 1	Bank	8,000

Purchases a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 3	Bank	2,500			

Sales a/c					
Date	Detail	Amount	Date	Detail	Amount
			Jan 20	T/R	3,400

Doyle - Trade receivables a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 20	Sales Rev.	3,400			

*A **trade receivable** account records the money owed from a customer for credit sales (the goods were given before the business received money for the goods)

Working example

* Jan 20 - Sold the goods **on credit** to Doyle for £3,400.

Jan 20	Dr.	Doyle (Trade receivables) (*Assets ↑)	3,400	
	Cr.	Sales revenue (*Revenue ↑)		3,400

Check the accounting equation:

Assets: Bank + Trade receivables = 5,500 + 3,400 = 8,900

Capital: (OC + NC – D) + R – E = 8,000 + 0 - 0 + 3,400 - 2,500 = 8,900

Liability: Nil

A (8,900) = C (8,900) + L (0)

Working example

* Jan 29 - Paid heat and light of £80 by cheque.

Bank a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 1	Cap. Intro.	8,000	Jan 3	Pur.	2,500
			Jan 29	H&L	80

Capital introduced a/c					
Date	Detail	Amount	Date	Detail	Amount
			Jan 1	Bank	8,000

Purchases a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 3	Bank	2,500			

Sales a/c					
Date	Detail	Amount	Date	Detail	Amount
			Jan 20	T/R	3,400

Doyle - Trade receivables a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 20	Sales Rev.	3,400			

Heat and light a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 29	Bank	80			

Working example

* Jan 29 - Paid heat and light of £80 by cheque.

Jan 29	Dr.	Heat and Light (*Expenses ↑)	80	
	Cr.	Bank (*Assets ↓)		80

Check the accounting equation:

Assets: **Bank** + Trade receivables = (5,500 - 80) + 3,400 = 8,820

Capital: (OC + NC – D) + R – **E** = 8,000 + 0 - 0 + 3,400 - (2,500 + 80) = 8,820

Liability: Nil

A (8,820) = C (8,820) + L (0)

Find the balance

- When the total amount of money on the debit side of an account is greater than that on the credit side, the account is said to have a **debit balance**.
When the reverse is the case, the account is said to have a **credit balance**.
- How much money does XYZ have in the bank at the end of January ?

Bank a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 1	Capital Introduced	8,000	Jan 3	Purchases	2,500
			Jan 29	Heat & Light	80
		5,420			

- Not a journal entry.
- Just calculation of the current balance after the above transactions
- record on the higher side
- To make trial balance (*Lecture 3*)

Exercise 1

Complete the ledger entry and journal entry for the following transactions

- Suzhou Ltd. bought a new van for £10,000 cash on 1 Jan

Cash a/c					
Date	Detail	Amount	Date	Detail	Amount
			Jan 1	Vehicles	10,000

Motor Vehicles a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 1	Cash	10,000			

Ledger (T-account) entry

Jan 1	Dr.	Motor vehicles	10,000	
	Cr.	Cash		10,000

Journal entry

Exercise 1

Finish the ledger entry and journal entry for the following transactions

- 5 Jan - Paid £250 in cash for expenses related to the van

Cash a/c					
Date	Detail	Amount	Date	Detail	Amount
			Jan 1	Vehicles	10,000
			Jan 5	Motor expense	250

Motor Vehicles a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 1	Cash	10,000			

Motor expenses a/c					
Date	Detail	Amount	Date	Detail	Amount
Jan 5	Cash	250			

Jan 5	Dr.	Motor expenses	250	
	Cr.	Cash		250

Ledger (T-account) entry

Journal entry

Exercise 2

Complete the journal entry for the following transactions.

- Lily is the owner of a sole trader business. On Feb 1, she bought office machinery for £1,000 in cash.

Feb 1	Dr.	Machinery	1,000	
	Cr.	Cash		1,000

- Lily bought a lorry for £500 by cheque on Feb

Feb 4	Dr.	Motor vehicles	500	
	Cr.	Bank		500

Exercise 2

Complete the journal entry for the following transactions.

- A loan of £400 was received by cheque from Earls and lodged to the business bank account on Feb 7.

Feb 7	Dr.	Bank	400	
	Cr.	Loan		400

- On Feb 9, she paid stationery (classified as an expense) for office use by cheque, £80.

Feb 9	Dr.	Stationary	80	
	Cr.	Bank		80

Exercise 2

Complete the journal entry for the following transactions.

- She withdrew the cash £200 from her business to buy a gift for her family on Feb 15.

Feb 15	Dr.	Drawings	200	
	Cr.	Cash		200

- On Feb 16, she paid rates by cash, £ 80.

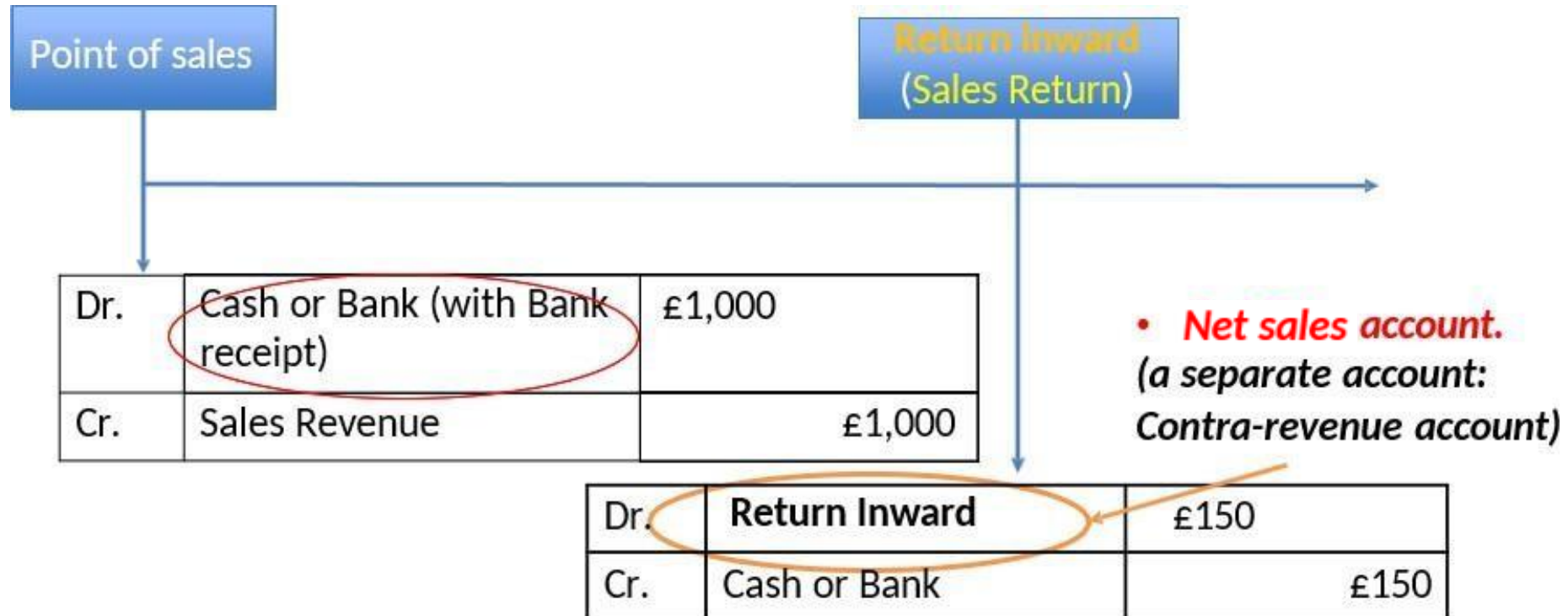
Feb 16	Dr.	Rate	80	
	Cr.	Cash		80

The nature of cash and credit transactions

- A *cash transaction* is one where goods or services are paid for in cash/cheque/bank transfer when they are received or delivered.
- A *credit transaction* is one where payment is made or received sometime **after** delivery (normally in one installment after a few weeks).
 - are extremely common in many industries.
 - Trade payable (L) & Trade receivable (A)

Sales transactions & sales return

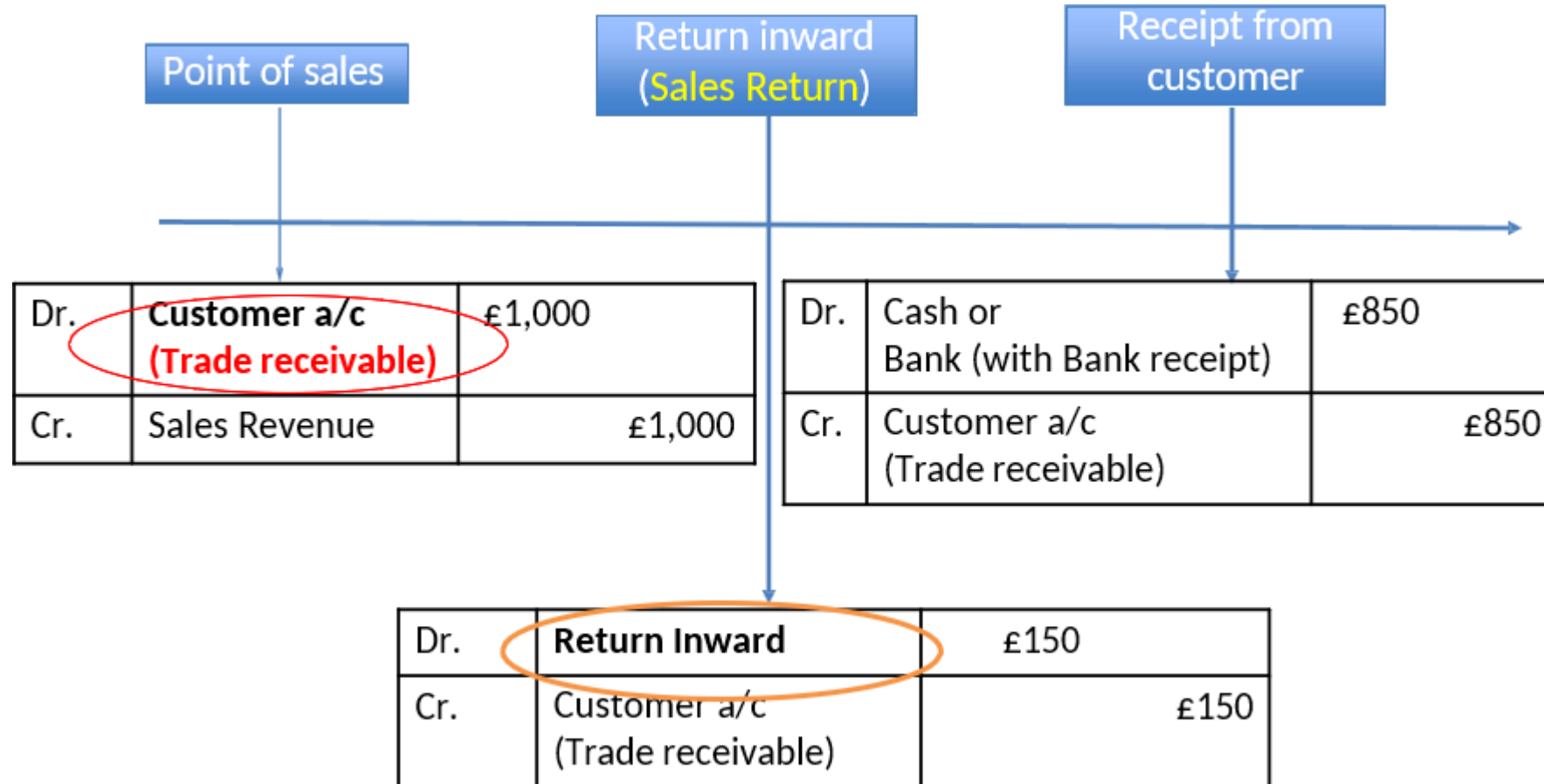
- Cash sales



* **Net sales rev. = £1,000 - £150 = £850**

Sales transactions & sales return

- Credit sales



*** Net sales rev. = £1,000 - £150 = £850**

Purchase transactions & purchase return

- Cash Purchase

** Purchase is an expense a/c*

Point of Purchase

Return outward
(Purchase return)

Dr.	Purchase	£1,000
Cr.	Cash or Bank (with Bank payment)	£1,000

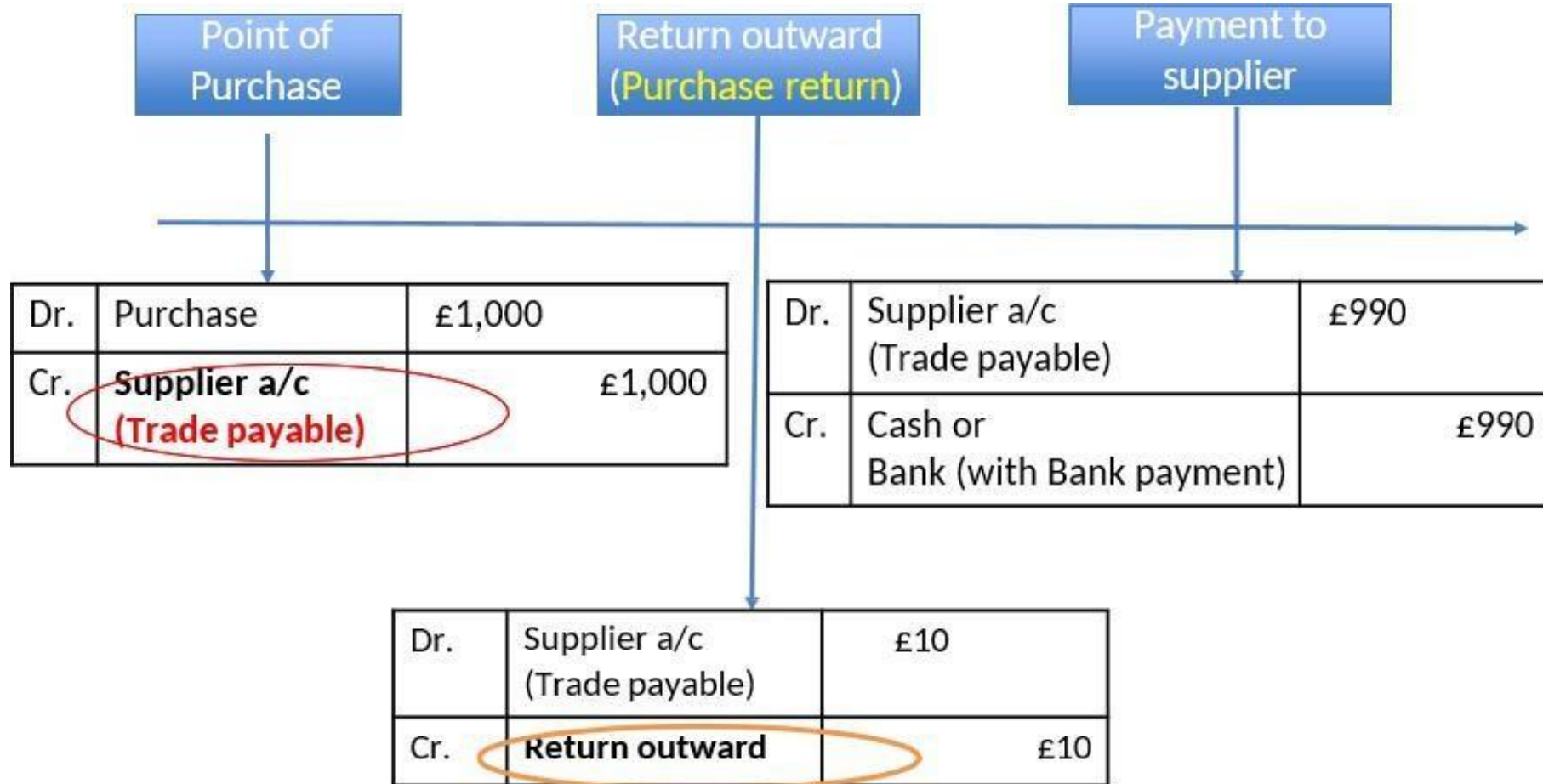
*• Net Purchase a/c.
(A separate a/c:
contra-purchase account)*

Dr.	Cash or Bank (with Bank payment)	£10
Cr.	Return outward	£10

** Net Purchase = £1,000 - £10 = £990*

Purchase transactions & purchase return

- Credit purchase



* Net Purchase = £1,000 - £10 = £990

Summary

- Remember these debit and credit rules for different types of accounts
 - The **increase** of **Assets/Expenses** is on the **Dr.** side and **Liabilities/Capital/Revenue** on the **Cr.** side.
 - The **decrease** of **Assets/Expenses** is on the **Cr.** side and **Liabilities/Capital/Revenue** on the **Dr.** side.

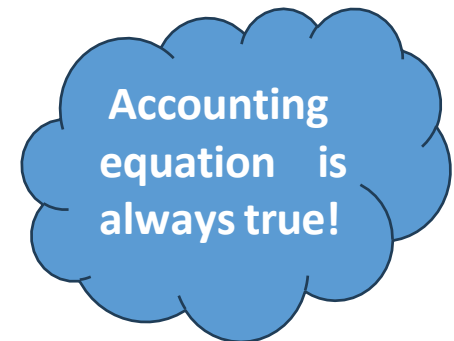
Assets	
Debit (+)	Credit(-)

Liabilities	
Debit (-)	Credit(+)

Capital	
Debit (-)	Credit(+)

Expenses	
Debit (+)	Credit(-)

Revenue	
Debit (-)	Credit(+)



Summary

- Transaction information can be expressed in 3 ways. All of them are **indicating the SAME THING!**

Narrative explanation

Jan 1 – The owner of the business put £1,000 cash into the business

Ledger entry

Cash a/c						Capital introduced a/c					
Date	Detail	Amount	Date	Detail	Amount	Date	Detail	Amount	Date	Detail	Amount
Jan 1	Cap. Intro.	1,000							Jan 1	Cash	1,000

Journal entry

Jan 1	Dr.	Cash	1,000	
	Cr.	Capital introduced		1,000

Homework

- ▶ Good review of the content of this lecture
- ▶ Attempt tutorial questions on time

Thanks for your listening

